CHAPTER B

# Fluency bank

Two questions per session, in 6 to 8 minutes including the correction. The identifiers allow revisits; they are not a list to be done in full at every lesson. The availability dates are based on the planned teaching, and should be shifted if the topic has not yet been studied.

### R001 – Fractions

Calculate $\frac{2}{3} + \frac{1}{4}$. **Solution.** $\frac{11}{12}$: common denominator 12.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R002 – Order of operations

Calculate $5 - 2(3 - 7)$. **Solution.** 13: brackets, product, difference.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R003 – Powers

Simplify $\frac{10^5 \times 10^{-3}}{10^4}$. **Solution.** $10^{-2} = 0.01$.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R004 – Roots

Simplify $\sqrt{50}$. **Solution.** $5\sqrt{2}$.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R005 – Proportion

18 represents 30% of a total. What is the total? **Solution.** $\frac{18}{0.3} = 60$.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R006 – Direct rate

A price of 80 falls by 15%. Final value? **Solution.** $80 \times 0.85 = 68$.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R007 – Initial value

After a 20% increase, the value is 72. Initial value? **Solution.** $\frac{72}{1.2} = 60$.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R008 – Calculated rate

A quantity goes from 40 to 50. Rate? **Solution.** $\frac{50-40}{40} = 25\%$.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R009 – Successive rates

A 20% increase then a 20% decrease: overall rate? **Solution.** $1.2 \times 0.8 = 0.96$: a 4% decrease.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R010 – Reciprocal rate

Which rate compensates a 20% decrease? **Solution.** $\frac{1}{0.8} = 1.25$: a 25% increase.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R011 – Zero product

Solve $(2x-1)(x+3) = 0$. **Solution.** $x = \frac{1}{2}$ or $x = -3$.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R012 – Expansion

Expand $(2x-3)^2$. **Solution.** $4x^2 - 12x + 9$.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R013 – Factorisation

Factorise $3x^2 - 12x$. **Solution.** $3x(x-4)$.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R014 – Sign of an affine function

Solve $-3x + 6 \geq 0$. **Solution.** $x \leq 2$: reverse the inequality when dividing by -3.
*Prerequisites:* Previous learning
*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R015 – Slope

Slope of the line through $(1, 4)$ and $(3, 0)$? **Solution.** $-2$.
*Prerequisites:* Previous learning
*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R016 – Line

Equation of the line of slope -2 through $(1, 4)$? **Solution.** $y = -2x + 6$.
*Prerequisites:* Previous learning
*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R017 – Weighted mean

The values 2 and 5 have counts 3 and 2. Mean? **Solution.** $\dfrac{6 + 10}{5} = 3.2$.
*Prerequisites:* Previous learning
*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R018 – Median

Median of 1, 2, 2, 5, 8, 10? **Solution.** $\dfrac{2 + 5}{2} = 3.5$ using the convention of the mean of the two central values.
*Prerequisites:* Previous learning
*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R019 – Quartiles

For the data set 1, 2, 3, 4, 5, 6, 7, 8, give $Q_1$ and $Q_3$ with the ceiling ranks $\left\lceil \dfrac{n}{4} \right\rceil$ and $\left\lceil \dfrac{3n}{4} \right\rceil$. **Solution.** $Q_1 = 2$, $Q_3 = 6$.
*Prerequisites:* Previous learning
*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R020 – Standard deviation of data

For 1 and 3 with equal weights, mean, variance and standard deviation? **Solution.** Mean 2, variance $\dfrac{[(1-2)^2 + (3-2)^2]}{2} = 1$, standard deviation 1.
*Prerequisites:* Previous learning
*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R021 – Units

Convert 1 min 40 s to seconds; 2400 frames over this duration give how many per second? **Solution.** 100 s; 24 frames/s.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R022 – Simple probability

With a fair die, probability of a multiple of 3? **Solution.** $\frac{2}{6} = \frac{1}{3}$.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R023 – Complement

$P(A) = \frac{3}{8}$. Give $P(\overline{A})$. **Solution.** $\frac{5}{8}$.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R024 – Interval

Write $x \geq -2$ and $x < 3$ as an interval. **Solution.** $[-2, 3)$.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R025 – Logic

Negate “all the students passed. **Solution.** “At least one student did not pass.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R026 – Histogram

Two classes of widths 2 and 4 have the same count 20. In a histogram with areas proportional to counts, what are the relative heights? **Solution.** Heights 10 and 5: the area, not the height alone, encodes the count.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R027 – Pie chart

A sector measures 72°. Proportion? **Solution.** $\frac{72}{360} = 20\%$.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R028 – Scale

An axis starts at 90. Bars stop at 100 and 110. Is the second value double the first? **Solution.** No: $\frac{110}{100} = 1.1$. The heights drawn above 90 make this comparison misleading.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R029 – Percentage of a percentage

40% of the students take A; among them, 25% take B. What proportion takes A and B? **Solution.** $0.4 \times 0.25 = 10\%$.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R030 – Proportion and count

A category represents 0.125 of 80 people. Count? **Solution.** 10 people.

*Prerequisites:* Previous learning

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.

### R031 – Defining and representing a sequence

Calculate the first four terms of $w_0 = 2$, $w_{n+1} = w_n + 2n + 1$; suggest an explicit formula. **Solution.** $2, 3, 6, 11$. The formula $w_n = n^2 + 2$ works: it gives the first term and $(n+1)^2 + 2 - (n^2 + 2) = 2n + 1$.

*Prerequisites:* Algebraic manipulation, integer powers, percentages, affine function, Python loop.

*Available from 2026-09-09; suggested revisits:* 2026-09-17, 2026-10-01, 2026-11-03.

### R032 – Arithmetic sequences

We know $u_3 = 17$ and $u_8 = 37$ for an arithmetic sequence. Determine $r$ and $u_0$. **Solution.** $37 - 17 = 5r$, so $r = 4$ and then $u_0 = 17 - 3 \times 4 = 5$.

*Prerequisites:* Algebraic manipulation, integer powers, percentages, affine function, Python loop.

*Available from 2026-09-11; suggested revisits:* 2026-09-22, 2026-10-06, 2026-11-03.

### R033 – Geometric sequences

Study the variation of $u_n = -2 \times 3^n$, then of $v_n = 2(-1)^n$. **Solution.** $u_{n+1} - u_n = -4 \times 3^n < 0$: decreasing. $v_n$ alternates between 2 and $-2$: neither increasing nor decreasing.

*Prerequisites:* Algebraic manipulation, integer powers, percentages, affine function, Python loop.

*Available from 2026-09-16; suggested revisits:* 2026-09-24, 2026-10-08, 2026-11-05.

### R034 – Sums, indices and proofs

Calculate $\sum_{k=2}^{5} 2^k$ without the summation symbol in the answer. **Solution.** $4+8+16+32=60$, or $\frac{4(1-2^4)}{1-2}=60$.

*Prerequisites:* Algebraic manipulation, integer powers, percentages, affine function, Python loop.

*Available from 2026-09-18; suggested revisits:* 2026-09-29, 2026-10-13, 2026-11-10.

### R035 – Conjecturing a limit and programming a threshold

For $u_n = 48(\frac{1}{2})^n$, determine the first indices where $u_n \leq 6$ and where $u_n < 6$. **Solution.** $u_3 = 6$: index 3 for $\leq 6$, index 4 for $< 6$.

*Prerequisites:* Algebraic manipulation, integer powers, percentages, affine function, Python loop.

*Available from 2026-09-18; suggested revisits:* 2026-09-29, 2026-10-13, 2026-11-10.

### R036 – Three forms, three readings

Give three forms of $g(x) = x^2 + 2x - 3$. Read off its roots and its minimum. **Solution.** $g(x) = (x+1)^2 - 4 = (x-1)(x+3)$. Roots 1 and $-3$, minimum $-4$ at $-1$.

*Prerequisites:* Special products, zero-product property, sign of a product, square roots.

*Available from 2026-09-23; suggested revisits:* 2026-10-01, 2026-10-15, 2026-11-12.

### R037 – Discriminant and solving

Solve $3x^2 - 6x + 3 = 0$ and $x^2 + 2x + 5 = 0$. **Solution.** The first is $3(x-1)^2 = 0$, so 1. For the second, $\Delta = -16 < 0$: no real solution.

*Prerequisites:* Special products, zero-product property, sign of a product, square roots.

*Available from 2026-09-25; suggested revisits:* 2026-10-06, 2026-11-03, 2026-11-17.

### R038 – Factorising without automating the discriminant

Determine the polynomial f of degree 2 such that $f(1) = f(4) = 0$ and $f(0) = -8$. **Solution.** $f(x) = a(x-1)(x-4)$; $4a = -8$, so $f(x) = -2(x-1)(x-4)$.

*Prerequisites:* Special products, zero-product property, sign of a product, square roots.

*Available from 2026-09-30; suggested revisits:* 2026-10-08, 2026-11-03, 2026-11-19.

### R039 – Sign, inequalities and optimisation

Solve $x^2 - 4x - 5 < 0$, then maximise $x(8-x)$ on $[0,3]$. **Solution.** $(x-5)(x+1) < 0$ gives $(-1,5)$. The vertex of $x(8-x)$ is at 4, outside the domain; on $[0,3]$ the function is increasing and its maximum is 15 at 3.

*Prerequisites:* Special products, zero-product property, sign of a product, square roots.

*Available from 2026-10-02; suggested revisits:* 2026-10-13, 2026-11-03, 2026-11-24.

### R040 – Projection and angle

Two vectors have norms 2 and 5 and form 90°, then 180°. Calculate the products. **Solution.** 0 then $-10$. The cosine is 0 and $-1$ respectively.

*Prerequisites:* Vectors, Chasles's relation, Pythagoras, cosine in a right triangle, orthonormal frame.

*Available from 2026-10-07; suggested revisits:* 2026-10-15, 2026-11-03, 2026-11-26.

### R041 – Coordinates, norm and orthogonality

For which real number t are the vectors $(t, 2)$ and $(3, -6)$ orthogonal? **Solution.** $3t - 12 = 0$, so $t = 4$.

*Prerequisites:* Vectors, Chasles's relation, Pythagoras, cosine in a right triangle, orthonormal frame.

*Available from 2026-10-09; suggested revisits:* 2026-11-03, 2026-11-03, 2026-12-01.

### R042 – Vector calculus and identities

Given $\|\vec{u}\| = 2$, $\|\vec{v}\| = 3$, $\vec{u} \cdot \vec{v} = 1$. Calculate $\|\vec{u} - \vec{v}\|$. **Solution.** The square is $4 - 2 + 9 = 11$; the norm is $\sqrt{11}$.

*Prerequisites:* Vectors, Chasles's relation, Pythagoras, cosine in a right triangle, orthonormal frame.

*Available from 2026-10-14; suggested revisits:* 2026-11-03, 2026-11-05, 2026-12-03.

### R043 – Al-Kashi and choosing a method

A triangle has $AB = AC = 4$ and $BC = 4\sqrt{2}$. Determine the angle A. **Solution.** $32 = 16 + 16 - 32\cos A$, so $\cos A = 0$ and $A = 90°$.

*Prerequisites:* Vectors, Chasles's relation, Pythagoras, cosine in a right triangle, orthonormal frame.

*Available from 2026-10-16; suggested revisits:* 2026-11-03, 2026-11-10, 2026-12-08.

### R044 – The circle with diameter AB

For $A(0,0)$, $B(6,0)$, determine the points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB} = 7$. **Solution.** $MI^2 - 9 = 7$: circle with centre $(3,0)$ and radius 4.

*Prerequisites:* Vectors, Chasles's relation, Pythagoras, cosine in a right triangle, orthonormal frame.

*Available from 2026-10-16; suggested revisits:* 2026-11-03, 2026-11-10, 2026-12-08.

### R045 – Rate of change and secants

Calculate the rate of $f(x) = x^2 + 3x$ between 2 and $2 + h$. **Solution.** $f(2+h) - f(2) = 7h + h^2$; rate $7 + h$ for $h \neq 0$.

*Prerequisites:* Images, slope of a line, algebraic manipulation, square root and absolute value.

*Available from 2026-11-04; suggested revisits:* 2026-11-12, 2026-11-26, 2027-01-05.

### R046 – Derivative at a point

For $f(x) = 3x^2 - 2x$, calculate $f\prime(1)$ from the rate. **Solution.** The rate is $4 + 3h$, so the derivative at 1 is 4.

*Prerequisites:* Images, slope of a line, algebraic manipulation, square root and absolute value.

*Available from 2026-11-06; suggested revisits:* 2026-11-17, 2026-12-01, 2027-01-05.

### R047 – Tangent and affine approximation

Given $f(3) = 5$ and $f\prime(3) = -2$. Write the tangent and approximate $f(3.04)$. **Solution.** $y = 5 - 2(x - 3) = -2x + 11$; approximation $5 - 2(0.04) = 4.92$.

*Prerequisites:* Images, slope of a line, algebraic manipulation, square root and absolute value.

*Available from 2026-11-11; suggested revisits:* 2026-11-19, 2026-12-03, 2027-01-05.

### R048 – When differentiability fails

Justify that $g(x) = |x - 2|$ is not differentiable at 2. Also study the differentiability of $x \mapsto \sqrt{x}$ at 0 through its rate. **Solution.** The rate at 2 is $\dfrac{|h|}{h}$, with two opposite limiting values. For the square root at 0, the rate $\dfrac{1}{\sqrt{h}}$ for $h > 0$ exceeds every bound as h gets closer to 0: no real derivative.

*Prerequisites:* Images, slope of a line, algebraic manipulation, square root and absolute value.

*Available from 2026-11-13; suggested revisits:* 2026-11-24, 2026-12-08, 2027-01-05.

### R049 – Conditioning: changing the denominator

Out of 60 objects, 20 are red; among the red ones, 5 are damaged. Calculate $P(R \cap D)$ and $P_R(D)$. **Solution.** $\dfrac{5}{60} = \dfrac{1}{12}$ and $\dfrac{5}{20} = \dfrac{1}{4}$.

*Prerequisites:* Fractions, proportions, probabilities on a finite sample space, intersection and union.

*Available from 2026-11-18; suggested revisits:* 2026-11-26, 2026-12-10, 2027-01-07.

### R050 – Trees, partitions and total probability

$P(A) = \dfrac{1}{3}$, $P_A(B) = \dfrac{3}{4}$, $P_{\overline{A}}(B) = \dfrac{1}{4}$. Calculate $P(B)$. **Solution.** $\dfrac{1}{4} + \dfrac{1}{6} = \dfrac{5}{12}$.

*Prerequisites:* Fractions, proportions, probabilities on a finite sample space, intersection and union.

*Available from 2026-11-20; suggested revisits:* 2026-12-01, 2026-12-15, 2027-01-12.

### R051 – Independence and disjointness

Given $P(A) = 0.4$, $P(B) = 0.5$, $P(A \cap B) = 0.18$. Are they independent? **Solution.** No: $0.4 \times 0.5 = 0.20 \neq 0.18$.

*Prerequisites:* Fractions, proportions, probabilities on a finite sample space, intersection and union.

*Available from 2026-11-25; suggested revisits:* 2026-12-03, 2026-12-17, 2027-01-19.

### R052 – Successive trials and Bernoulli trials

Four independent trials, each with probability $p = \frac{1}{3}$. Calculate the probability of four failures, then of at least one success. **Solution.** $(\frac{2}{3})^4 = \frac{16}{81}$ then $1 - \frac{16}{81} = \frac{65}{81}$.

*Prerequisites:* Fractions, proportions, probabilities on a finite sample space, intersection and union.

*Available from 2026-11-27; suggested revisits:* 2026-12-08, 2027-01-05, 2027-01-19.

### R053 – Simulating and discussing the model

A simulation of 2000 points gives 675 points below $y = x^2$. What area does it estimate? **Solution.** $\frac{675}{2000} = 0.3375$; this is an estimate, not the exact value.

*Prerequisites:* Fractions, proportions, probabilities on a finite sample space, intersection and union.

*Available from 2026-11-27; suggested revisits:* 2026-12-08, 2027-01-05, 2027-01-19.

### R054 – Derivative function and standard functions

Differentiate $x^5$, $x^{-2}$ and $\sqrt{x}$, stating the domains. **Solution.** $5x^4$ on $\mathbb{R}$; $-\frac{2}{x^3}$ for $x \neq 0$; $\frac{1}{2\sqrt{x}}$ for $x > 0$.

*Prerequisites:* Derivative at a point, algebraic calculation, sign of a quadratic, intervals.

*Available from 2026-12-02; suggested revisits:* 2026-12-10, 2027-01-05, 2027-01-21.

### R055 – Sum, product, reciprocal and quotient

Differentiate $(x^2+1)(3x-2)$, then $\frac{1}{x^2+1}$. **Solution.** $2x(3x-2) + 3(x^2+1) = 9x^2 - 4x + 3$; $-\frac{2x}{(x^2+1)^2}$, on $\mathbb{R}$.

*Prerequisites:* Derivative at a point, algebraic calculation, sign of a quadratic, intervals.

*Available from 2026-12-04; suggested revisits:* 2026-12-15, 2027-01-05, 2027-01-26.

### R056 – Sign of the derivative and extrema

Study $f(x) = x^2 - 4x + 1$ on $[0, 3]$ and give its extrema on this interval. **Solution.** $f' = 2x - 4$, so decreasing up to 2 then increasing. Minimum $-3$ at 2; values at the endpoints 1 and $-2$, so maximum 1 at 0.

*Prerequisites:* Derivative at a point, algebraic calculation, sign of a quadratic, intervals.

*Available from 2026-12-09; suggested revisits:* 2026-12-17, 2027-01-05, 2027-01-28.

### R057 – Parity, inequalities and comparison

Study the parity of $x^4 + x^2$ and $x^3 - x$; prove $x^2 + 1 \geq 2x$. **Solution.** First even, second odd; difference $x^2 + 1 - 2x = (x-1)^2 \geq 0$, equality at 1.

*Prerequisites:* Derivative at a point, algebraic calculation, sign of a quadratic, intervals.

*Available from 2026-12-11; suggested revisits:* 2027-01-05, 2027-01-05, 2027-02-02.

### R058 – Optimisation and Newton's method

A rectangle of perimeter 12 m has width x. Determine the maximum area by differentiation. **Solution.** $A(x) = x(6-x)$ on $[0,6]$; $A\prime = 6-2x$. Maximum 9 $\mathrm{m}^2$ at 3.

*Prerequisites:* Derivative at a point, algebraic calculation, sign of a quadratic, intervals.

*Available from 2026-12-11; suggested revisits:* 2027-01-05, 2027-01-05, 2027-02-02.

### R059 – Normal vector and Cartesian equation

Equation of the line through $(2,-1)$ with normal $(3,4)$. **Solution.** $3(x-2)+4(y+1)=0$, that is $3x+4y-2=0$.

*Prerequisites:* Vector coordinates, scalar product, linear systems, completing the square. The frame is always orthonormal.

*Available from 2027-01-06; suggested revisits:* 2027-01-19, 2027-01-28, 2027-03-02.

### R060 – Orthogonal projection and distance

Project $P(4,1)$ onto $d : x = 2$. **Solution.** $H(2,1)$ and distance 2; the normal is horizontal.

*Prerequisites:* Vector coordinates, scalar product, linear systems, completing the square. The frame is always orthonormal.

*Available from 2027-01-08; suggested revisits:* 2027-01-19, 2027-02-02, 2027-03-02.

### R061 – Equation of a circle

Recognise $x^2+y^2+2x-8y+17=0$. **Solution.** $(x+1)^2+(y-4)^2=0$: the single point $(-1,4)$, not a circle of strictly positive radius.

*Prerequisites:* Vector coordinates, scalar product, linear systems, completing the square. The frame is always orthonormal.

*Available from 2027-01-13; suggested revisits:* 2027-01-21, 2027-02-04, 2027-03-04.

### R062 – Solving a configuration

Intersect $(x-1)^2+(y+2)^2=9$ with $y=1$. **Solution.** $(x-1)^2=0$, so the only point is $(1,1)$; the line is tangent to the circle.

*Prerequisites:* Vector coordinates, scalar product, linear systems, completing the square. The frame is always orthonormal.

*Available from 2027-01-15; suggested revisits:* 2027-01-26, 2027-02-09, 2027-03-09.

### R063 – Variable, event and distribution

Two fair coins are tossed. X counts the heads. Write its distribution and $P(X \leq 1)$. **Solution.** Values 0, 1, 2; probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$. $P(X \leq 1) = \frac{3}{4}$.

*Prerequisites:* Probability, weighted mean, squares, trees and lists.

*Available from 2027-01-20; suggested revisits:* 2027-01-28, 2027-02-11, 2027-03-11.

### R064 – Expectation and linearity

A gross gain is 10 with probability $\frac{1}{5}$ and 0 otherwise. What stake makes the game fair? **Solution.** $E(G) = \frac{10}{5} = 2$. For a stake m, $E(G - m) = 2 - m$; fair stake 2.

*Prerequisites:* Probability, weighted mean, squares, trees and lists.

*Available from 2027-01-22; suggested revisits:* 2027-02-02, 2027-03-02, 2027-03-16.

### R065 – Variance, standard deviation and König–Huygens

For X with values $-1, 1$ equally likely, calculate the expectation, variance and standard deviation. **Solution.** $E(X) = 0$, $E(X^2) = 1$, variance 1 and standard deviation 1.

*Prerequisites:* Probability, weighted mean, squares, trees and lists.

*Available from 2027-01-27; suggested revisits:* 2027-02-04, 2027-03-02, 2027-03-18.

### R066 – Transforming and comparing gains

$E(X) = 3$, $V(X) = 4$. Give the characteristics of $Y = -2X + 5$. **Solution.** $E(Y) = -1$, $V(Y) = 16$, $\sigma(Y) = 4$.

*Prerequisites:* Probability, weighted mean, squares, trees and lists.

*Available from 2027-01-29; suggested revisits:* 2027-02-09, 2027-03-02, 2027-03-23.

### R067 – Samples and observed mean

For $\mu = 5$, $\sigma = 3$ and n=36, give the experimental interval asked for. **Solution.** $\frac{2\sigma}{\sqrt{n}} = 1$; interval $[4, 6]$.

*Prerequisites:* Probability, weighted mean, squares, trees and lists.

*Available from 2027-01-29; suggested revisits:* 2027-02-09, 2027-03-02, 2027-03-23.

### R068 – Defining a new function

What is the equation of the tangent to $e^x$ at 1? **Solution.** $y = e + e(x - 1) = ex$.

*Prerequisites:* Powers, geometric sequences, differentiation and sign.

*Available from 2027-02-03; suggested revisits:* 2027-02-11, 2027-03-02, 2027-03-25.

### R069 – Algebraic properties and comparisons

Simplify $\frac{e^{3x}}{e^{x-1}}$; solve $e^{x+2} > e^{2x}$. **Solution.** $e^{2x+1}$; $x + 2 > 2x$, so $x < 2$.

*Prerequisites:* Powers, geometric sequences, differentiation and sign.

*Available from 2027-02-05; suggested revisits:* 2027-03-02, 2027-03-02, 2027-03-30.

### R070 – Differentiating and studying

Differentiate $g(t) = 3e^{-2t}$ and study its direction of variation. **Solution.** $g'(t) = -6e^{-2t} < 0$ on $\mathbb{R}$: strictly decreasing.

*Prerequisites:* Powers, geometric sequences, differentiation and sign.

*Available from 2027-02-10; suggested revisits:* 2027-03-02, 2027-03-04, 2027-04-06.

### R071 – From discrete to continuous

For $Q(t) = 5e^{0.3t}$, express $Q(n)$ and $\dfrac{Q(t+2)}{Q(t)}$. **Solution.** $u_n = 5(e^{0.3})^n$; the ratio over two units is $e^{0.6}$.

*Prerequisites:* Powers, geometric sequences, differentiation and sign.

*Available from 2027-02-12; suggested revisits:* 2027-03-02, 2027-03-09, 2027-04-06.

### R072 – Building numerically and proving an inequality

Show that $e^{2x} \geq 1 + 2x$ for every real number x. **Solution.** Apply the inequality proved with the argument $2x$; equality only at x=0.

*Prerequisites:* Powers, geometric sequences, differentiation and sign.

*Available from 2027-02-12; suggested revisits:* 2027-03-02, 2027-03-09, 2027-04-06.

### R073 – Measuring an angle in radians

Convert 45°, 120°; give the arc for radius 2 m and angle $\dfrac{3\pi}{4}$. **Solution.** $\dfrac{\pi}{4}$, $\dfrac{2\pi}{3}$; length $\dfrac{3\pi}{2}$ m.

*Prerequisites:* Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.

*Available from 2027-03-03; suggested revisits:* 2027-03-11, 2027-03-25, 2027-04-27.

### R074 – Wrapping and the image of a real number

Place the images of $\dfrac{7\pi}{2}$ and $-\dfrac{5\pi}{3}$ by reducing them to within one turn. **Solution.** $\dfrac{7\pi}{2}$ has the image of $\dfrac{3\pi}{2}$; $-\dfrac{5\pi}{3}$ that of $\dfrac{\pi}{3}$.

*Prerequisites:* Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.

*Available from 2027-03-05; suggested revisits:* 2027-03-16, 2027-03-30, 2027-04-27.

### R075 – Sine, cosine and standard values

If $\cos x = \dfrac{3}{5}$ and the image of x is in the fourth quadrant, give $\sin x$. **Solution.** $\sin^2 x = \dfrac{16}{25}$; the sine is negative, so $-\dfrac{4}{5}$.

*Prerequisites:* Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.

*Available from 2027-03-10; suggested revisits:* 2027-03-18, 2027-04-06, 2027-04-29.

### R076 – Related angles and symmetries

Calculate the two coordinates of the images of $\frac{4\pi}{3}$ and $\frac{7\pi}{4}$. **Solution.** $(-\frac{1}{2}, -\frac{\sqrt{3}}{2})$ and $(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2})$.

*Prerequisites:* Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.

*Available from 2027-03-12; suggested revisits:* 2027-03-23, 2027-04-06, 2027-05-04.

### R077 – Approximating pi and putting it to use

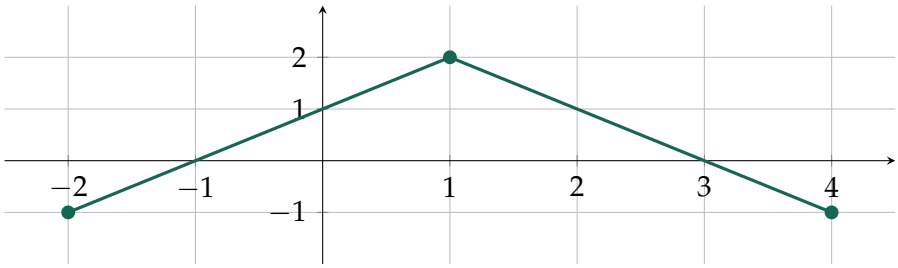
For n=6 and c=1, calculate the new chord after doubling. **Solution.** $h = \frac{\sqrt{3}}{2}$; new chord $\frac{1}{\sqrt{2+\sqrt{3}}} = \sqrt{2-\sqrt{3}}$.

*Prerequisites:* Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.

*Available from 2027-03-12; suggested revisits:* 2027-03-23, 2027-04-06, 2027-05-04.

### R078 – Reading a graph

On the figure R078, read $f(1)$, solve $f(x) = 0$ and $f(x) > 0$.




**Solution.** $f(1) = 2$; zeros $-1$ and 3; positive sign on $(-1, 3)$.

*Prerequisites:* Plotting points

*Available from 2026-09-03; suggested revisits:* 2026-09-10, 2026-09-24, 2026-11-03.